IOWA

Calculating PUF-PAS Sampling Rates and Effective Volumes in RStudio

A Step-by-Step Workflow | PUF-PAS Effective Volume Model

Version 2.1 - October 2026

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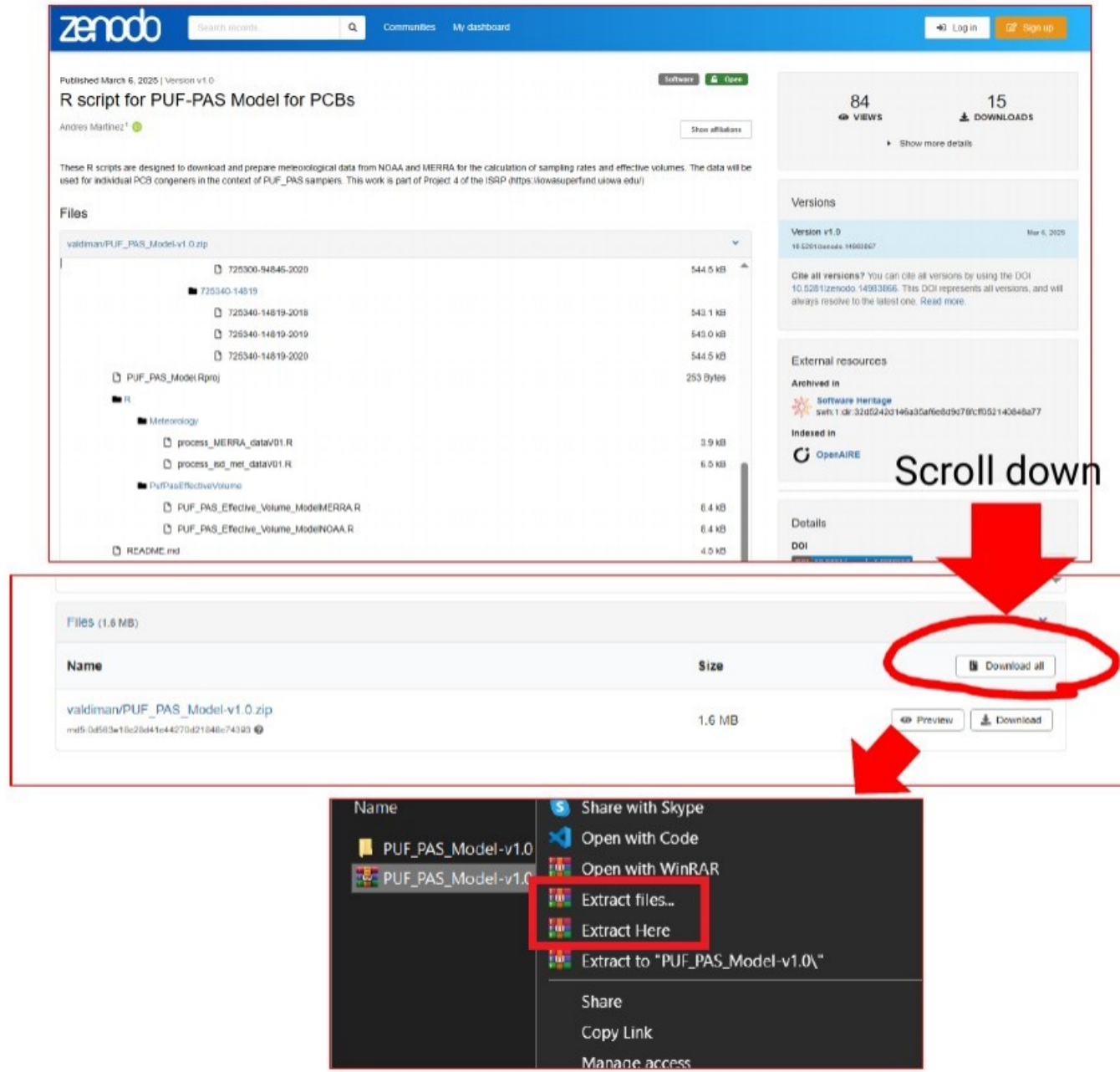
IIHR-Hydroscience & Engineering | University of Iowa

R-project Contents

Item	Function
PUF_PAS_Model.Rproj	Opens the project with the correct working directory
Subfolders.R	Creates the required folder structure
R/Meteorology	Contains NOAA and MERRA meteorological scripts
R/PufPasEffectiveVolume	Contains NOAA and MERRA effective-volume models
Data	Contains PCB physicochemical-property tables
Output	Stores meteorological data and model results
README.md	Basic project instructions

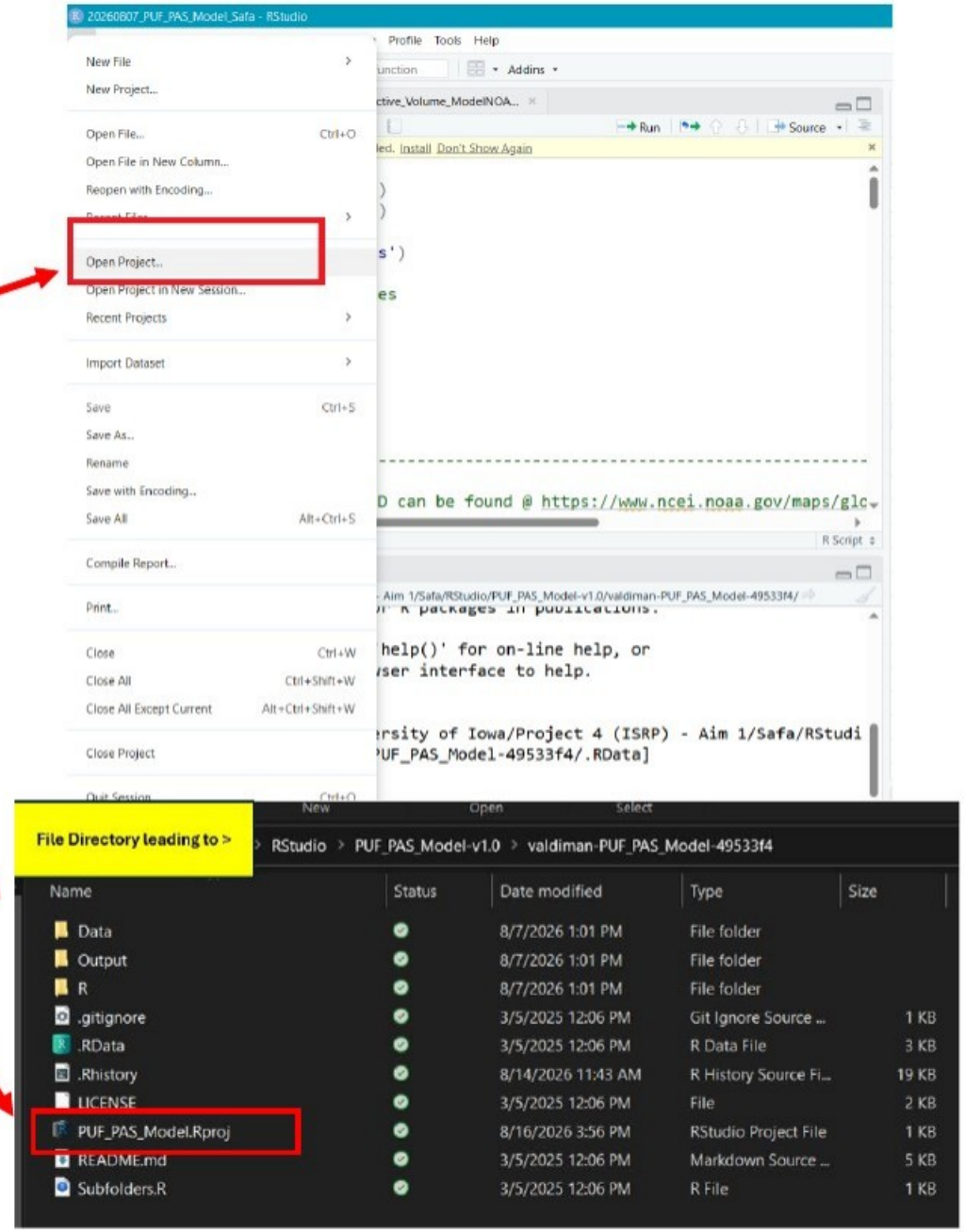
Download and Extract the Project

- **Prerequisite:** Make sure R and RStudio Desktop are installed on your computer before starting
- Go to the **Zenodo repository**: https://zenodo.org/records/14983867
- Download the complete ZIP file to your computer
- Locate the downloaded ZIP file, right-click it, and select Extract All...
- Choose a location for the extracted folder and complete the extraction. You may rename the extracted folder for convinience.
- **Important**: always work from the extracted folder. Do not open or run the R scripts directly from inside the ZIP archive.
- Open RStudio Desktop to begin working with the model files



Open the Project in Rstudio

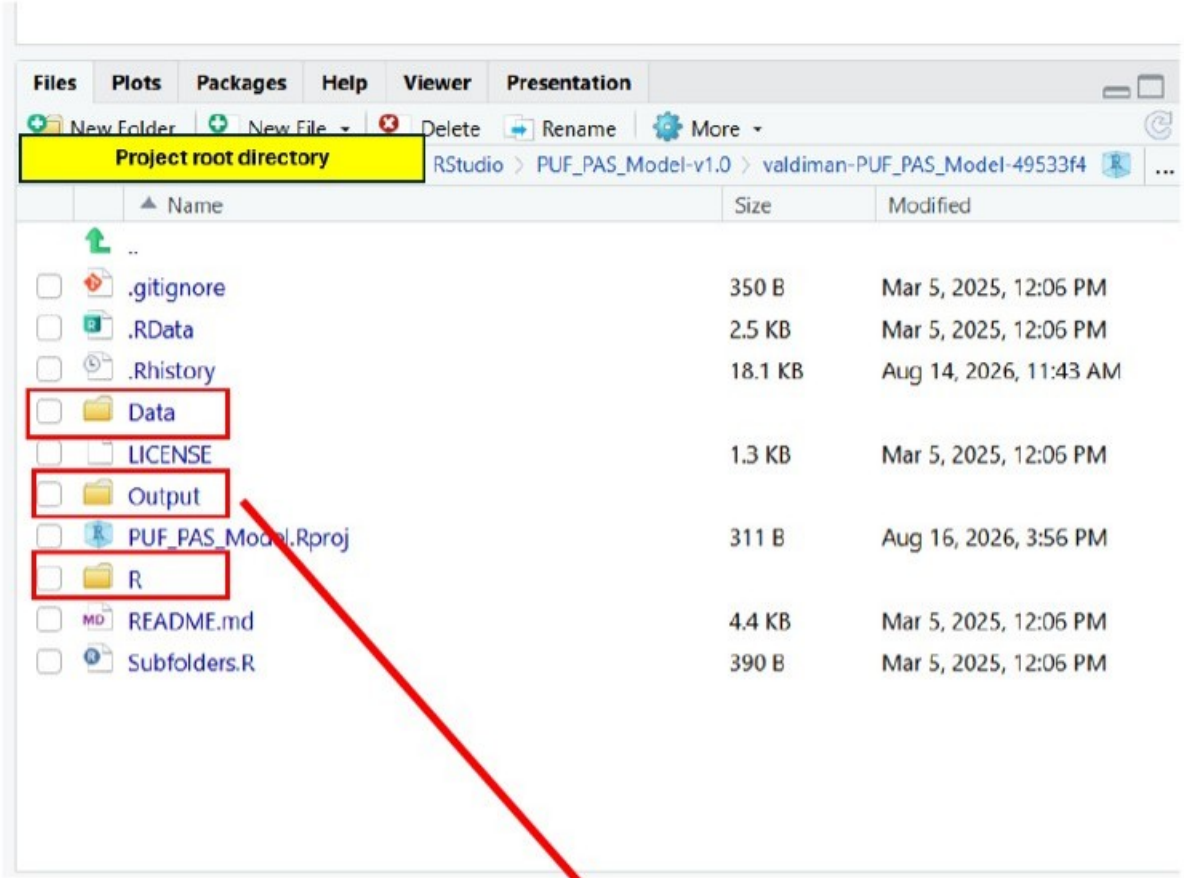
- Open RStudio Desktop
- **Select File → Open Project…**
- Navigate to the extracted project folder
- Select and open **PUF_PAS_Model.Rproj**
- The project will open in RStudio, with the project folder set as the working directory



Confirm that the Project Opened Correctly

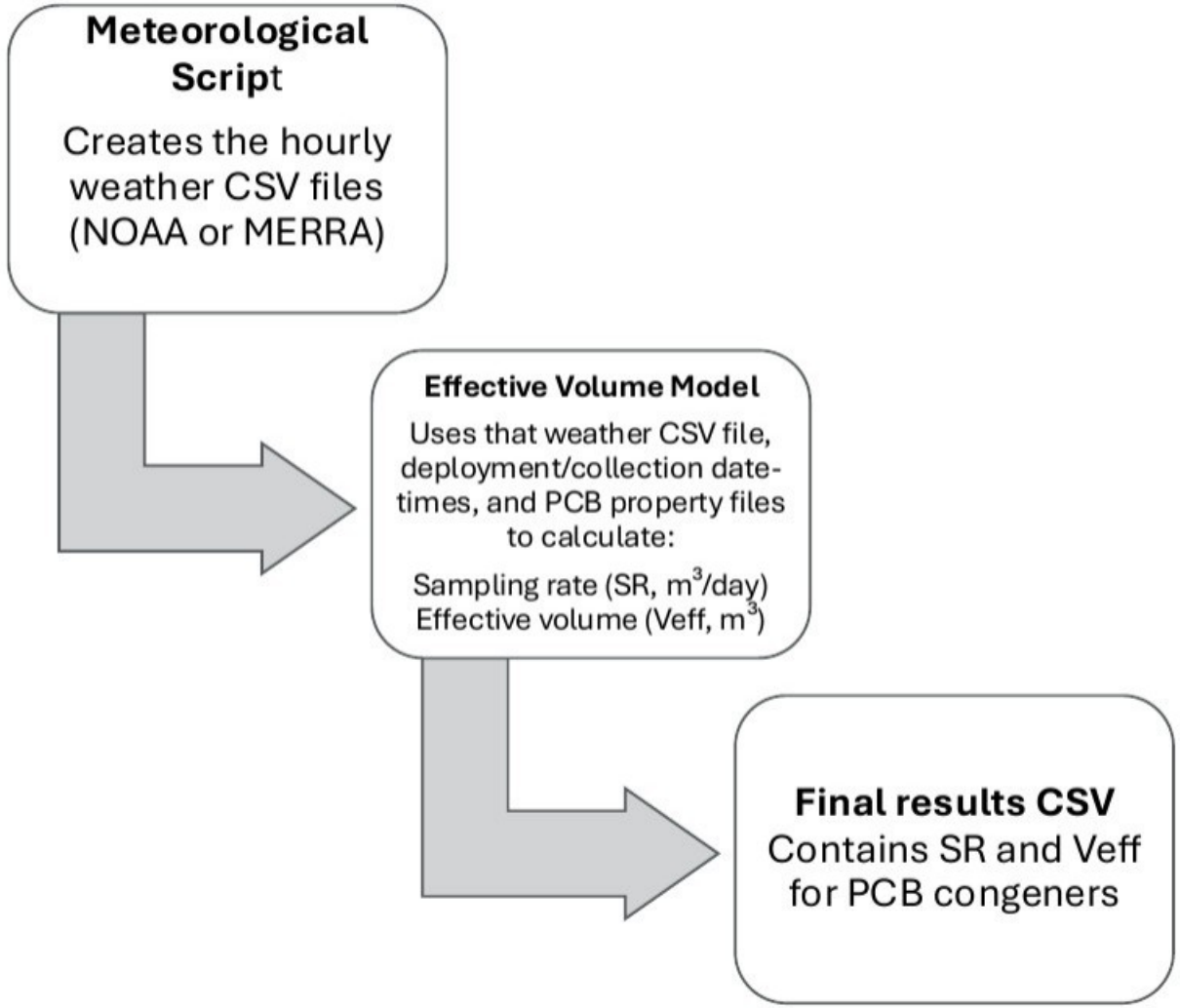
- The **Files** pane should display the project root containing "**`Data`**", "**`Output`**", and "**`R`**"

Item	Purpose
Data	PCB physicochemical-property input files
Output	Meteorological files and final model results
R	Meteorological-processing and effective-volume scripts
README.md	Project instructions
Subfolders.R	Creates the required folders if they are missing



Note: The CSV files already stored in Output are examples. Delete these example CSVs before running the scripts, but keep the Output folder and all its subfolders

Workflow


Meteorological Script
Creates the hourly weather CSV files (NOAA or MERRA)
Effective Volume Model
Uses that weather CSV file, deployment/collection date-times, and PCB property files to calculate:
Sampling rate (SR, m^3/day)
Effective volume (Veff, m^3)
Final results CSV
Contains SR and Veff for PCB congeners

Choose GHCNh or MERRA and gather required inputs

USE NOAA GHCNh

- A representative nearby station is available.
- The station record covers the deployment period.
- Required inputs: station ID, start year, end year.
- For incomplete years, V02 stops at the last available observation.

USE MERRA-2

- No suitable nearby NOAA station is available.
- The station record does not cover the sampling period.
- Required inputs: latitude, longitude, start and end dates.
- All deployment and collection times use UTC.

Find GHCNh stations: ncei.noaa.gov/access/search/datasets/global-historical-climatology-network-hourly/

Open the NOAA GHCNh or MERRA meteorological script

NOAA GHCNh

```
R/Meteorology/
process_GHCNh_dataV02.R
```

Select a station ID and a start/end year. V02 selects representative observations near hourly boundaries.

MERRA-2

```
R/Meteorology/
process_MERRA_dataV01.R
```

Use the existing MERRA workflow for site coordinates and the required date range.

NOAA GHCNh: enter station ID and years

Open process_GHCNh_dataV02.R and edit only the station and year settings:

```
# Chicago O'Hare International Airport
metadataID <- "USW00094846"
start_year<-2025
end_year<-2025
```

V02 does not use period_mode, end_month, or end_day. For an incomplete year, the output ends at the last available selected observation.

Run the meteorological script (NOAA GHCNh)

- Save the edited V02 script and click Source in RStudio.
- Review the number and percentage of missing hours for TA, TD, Pr, WS, WD, and QV.
- Review warnings for any variable with more than 72 consecutive missing hours.
- Missing values are forward-filled; leading gaps are backward-filled.
- QV is recalculated from filled dew-point temperature and pressure.
- Confirm the station, selected data period, and final saved-file message.

Quality warnings flag data limitations; they do not stop the filling procedure.

Locate the processed meteorological file (NOAA GHCNh)

Find the single model-ready CSV under:

```
Output/Data/GHCNh/<Station_ID>/
```

Example - Chicago O'Hare, 2025:

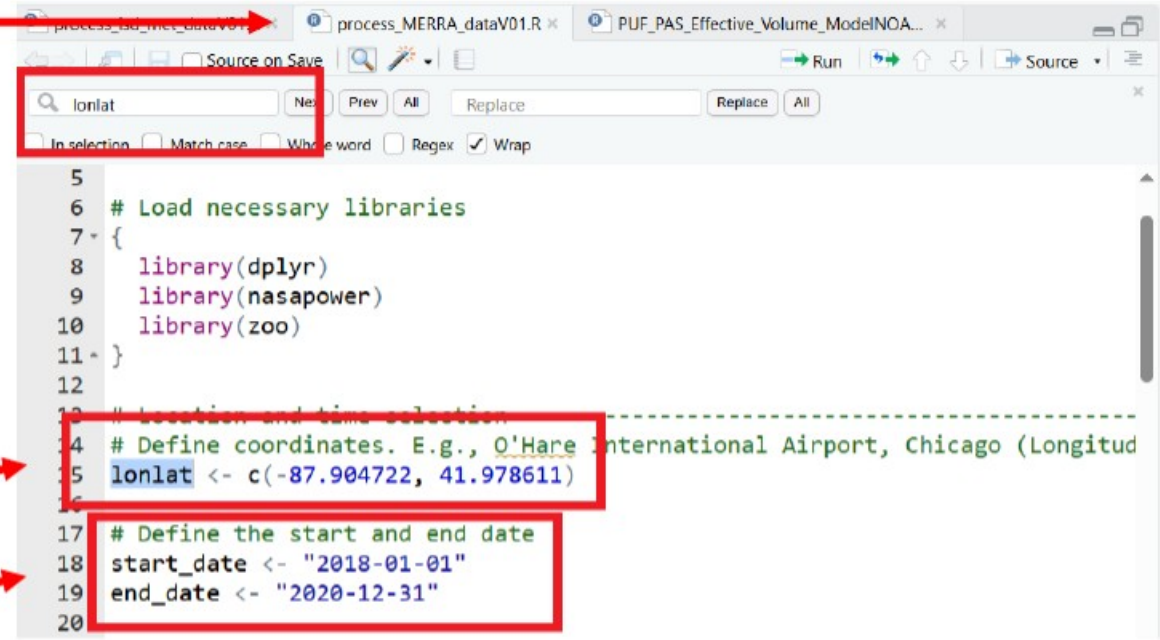
```
USW00094846-2025-2025-GHCNh-filled.csv
```

CSV columns: date (UTC), TA (K), Pr (Pa), WS (m/s), WD (degrees), QV (kg/kg). Use this file as met_data_path in the effective-volume script.

MERRA Workflow: Enter the Coordinates and Dates

- Open: "**process_MERRA_dataV01.R**"
- Install the required R packages if they are not already installed (lines 2 – 4)
- Load the required libraries before running the data-processing code (lines 8 – 10)
- Press **Ctrl + F** and search for "**lonlat**" (line 15)
- Replace the example coordinates with the coordinates of your sampling location. Enter the coordinates in longitude, latitude order. The coordinates can be obtained using **Google Earth**
- Replace the sampling years (yyyy-mm-dd)
- Search for **output_dir** (line 102) and replace the example site name in the output folder, and search for the **output_file_path** (line 108) and replace CSV filename (example: OHareAirport-2018-2020.csv)

Variable	Replace with
lonlat	Sampling-site longitude and latitude
start_date	First meteorological date required
end_date	Last meteorological date required
output_dir	Site-specific output folder name
output_file_path	Site- and period-specific CSV filename





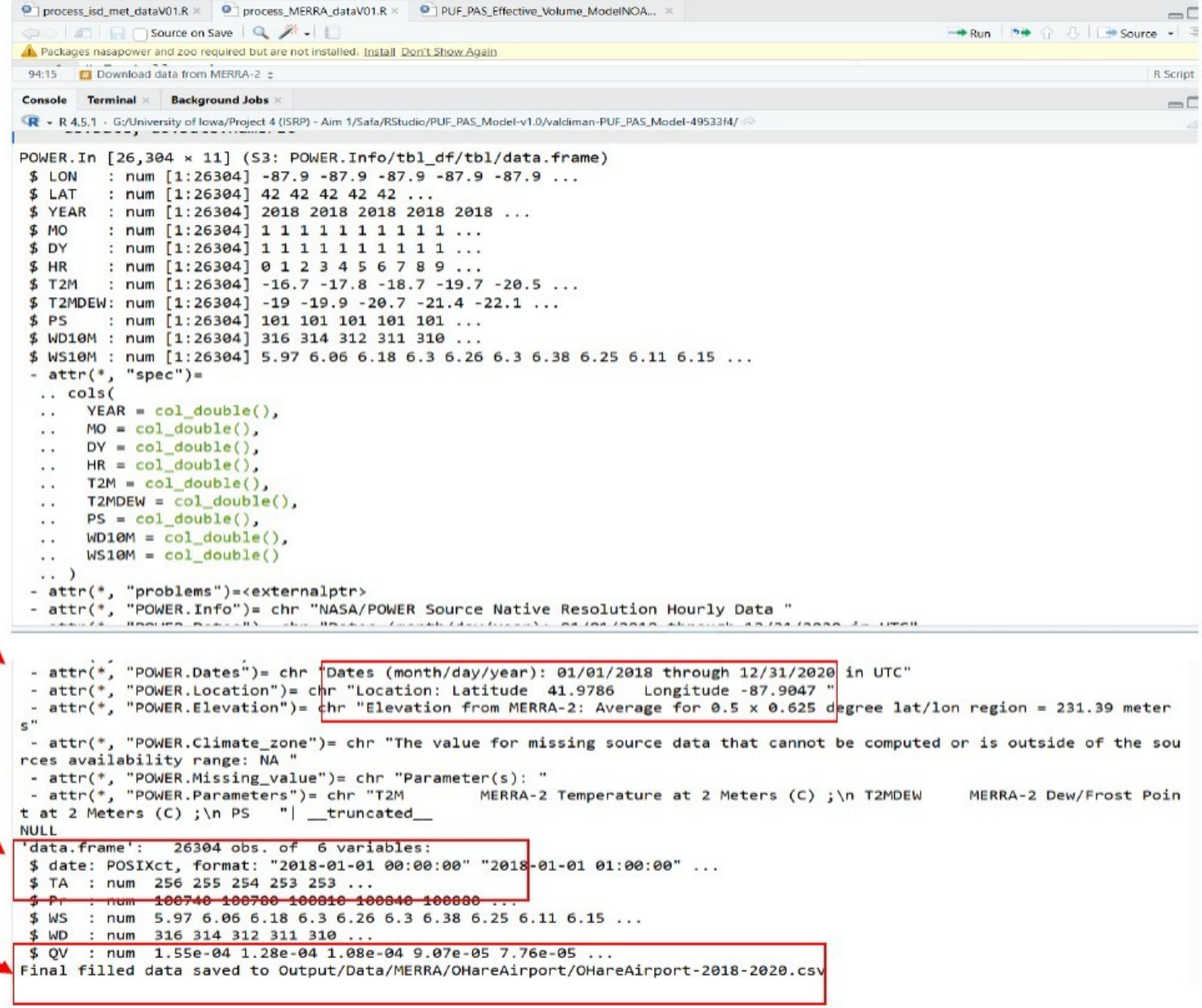
Run the MERRA Meteorological Script

- **To run the code:**
- If packages and libraries were already installed and loaded, no need to run it again (see instruction on previous slide)

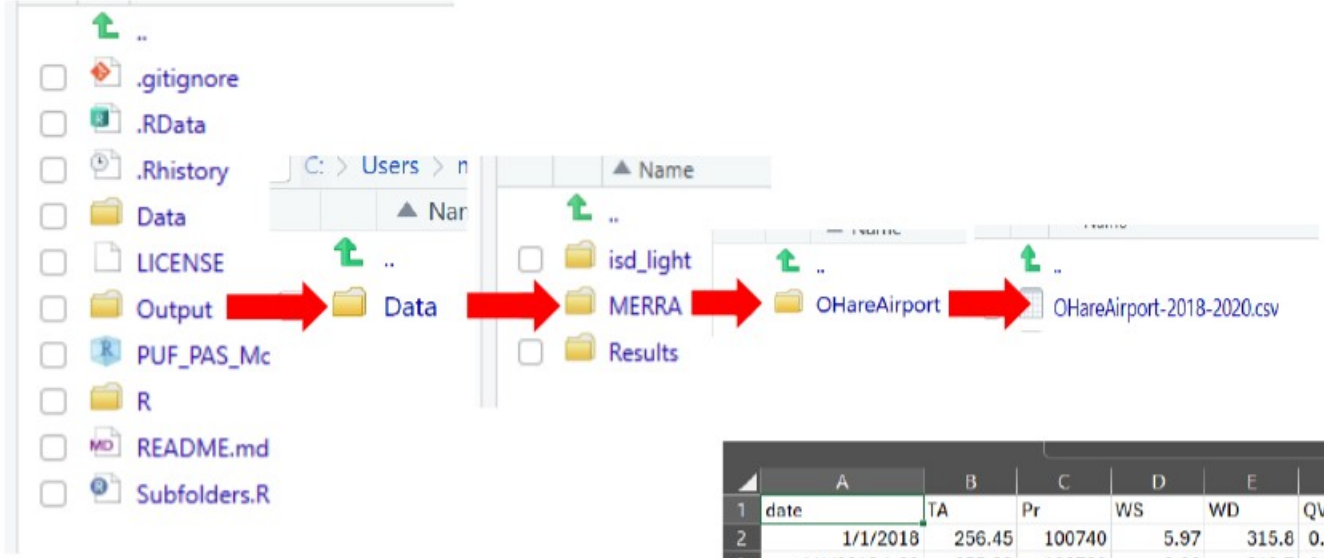
- Confirm the selected dates and coordinates in the **POWER summary**

- For this O'Hare 2018–2020 example, the processed dataset contains 26,304 hourly observations and six required variables

- Confirm completion using the **final saved-file** message



Locate the Processed Meteorological file (MERRA)



- In RStudio's Files pane, navigate to (MERRA example): **Output → Data → MERRA → OHareAirport**
- Locate the processed meteorological data file. In our example: OHareAirport-2018-2020.csv
- This CSV contains the processed hourly MERRA-2 meteorological data for the O'Hare 2018–2020 example
- Note the location where the file was saved, as its file path will be entered later as "**met_data_path**" in the MERRA effective-volume model
- MERRA filenames generally follow: SamplingLocation-StartYear-EndYear.csv; For example, "OHareAirport-2018-2020.csv"

Processed MERRA-2 CSV opened in Excel

	A	B	C	D	E	F
1	date	TA	Pr	WS	WD	QV
2	1/1/2018	256.45	100740	5.97	315.8	0.000155
3	1/1/2018 1:00	255.39	100780	6.06	313.7	0.000128
4	1/1/2018 2:00	254.44	100810	6.18	312	0.000108
5	1/1/2018 3:00	253.49	100840	6.3	310.8	9.07E-05
6	1/1/2018 4:00	252.65	100880	6.26	309.8	7.76E-05
7	1/1/2018 5:00	251.98	100910	6.3	310.7	6.77E-05
8	1/1/2018 6:00	251.54	100930	6.38	311.3	6.18E-05
9	1/1/2018 7:00	251.13	100970	6.25	310.7	5.75E-05
10	1/1/2018 8:00	250.76	100990	6.11	308	5.43E-05
11	1/1/2018 9:00	250.28	100990	6.15	305	5.05E-05
12	1/1/2018 10:00	249.85	100990	6.25	302.1	4.74E-05
13	1/1/2018 11:00	249.55	101020	6.34	299.4	4.54E-05
14	1/1/2018 12:00	249.51	101060	6.5	299.2	4.51E-05
15	1/1/2018 13:00	249.92	101110	6.7	301.7	4.79E-05
16	1/1/2018 14:00	251.3	101150	7.25	305.5	5.97E-05
17	1/1/2018 15:00	252.58	101190	7.18	307.8	7.30E-05
18	1/1/2018 16:00	253.97	101190	6.82	307.8	9.14E-05
19	1/1/2018 17:00	255.67	101140	6.47	307.1	0.00012
20	1/1/2018 18:00	257	101070	6.15	303.9	0.000151
21	1/1/2018 19:00	257.88	101040	5.98	301.3	0.000176
22	1/1/2018 20:00	258.22	101050	5.83	299.6	0.000188
23	1/1/2018 21:00	257.62	101070	5.44	297.2	0.000176
24	1/1/2018 22:00	255.52	101090	4.74	292.3	0.000132
25	1/1/2018 23:00	254.1	101120	4.57	285.5	0.00011
26	1/2/2018	253.25	101130	4.58	280.6	9.90E-05

OHareAirport-2018-2020

Open the Effective Volume Model Script (NOAA GHCNh or MERRA)

In RStudio, go to the Files pane → R → PufPasEffectiveVolume.

Select the appropriate model script:

NOAA GHCNh

PUF_PAS_Effective_Volume_ModelGHCNh.R

MERRA

PUF_PAS_Effective_Volume_ModelMERRA.R

Double-click the appropriate script to open it in the RStudio Source pane.

The effective-volume model uses the processed meteorological CSV, deployment/collection date-times, and the PCB property files to calculate congener-specific sampling rates and effective volumes.

NOAA GHCNh effective-volume model script

Open R/PufPasEffectiveVolume/PUF_PAS_Effective_Volume_ModelGHCNh.R

```
start_date = "2025-03-01 01:00:00"
end_date   = "2025-04-10 01:00:00"

met_data_path =
  "Output/Data/GHCNh/USW00094846/USW00094846-2025-2025-GHCNh-filled.csv"
```

Set pcb_properties_path and pcb_ppLFER_path to the PCB input CSVs.

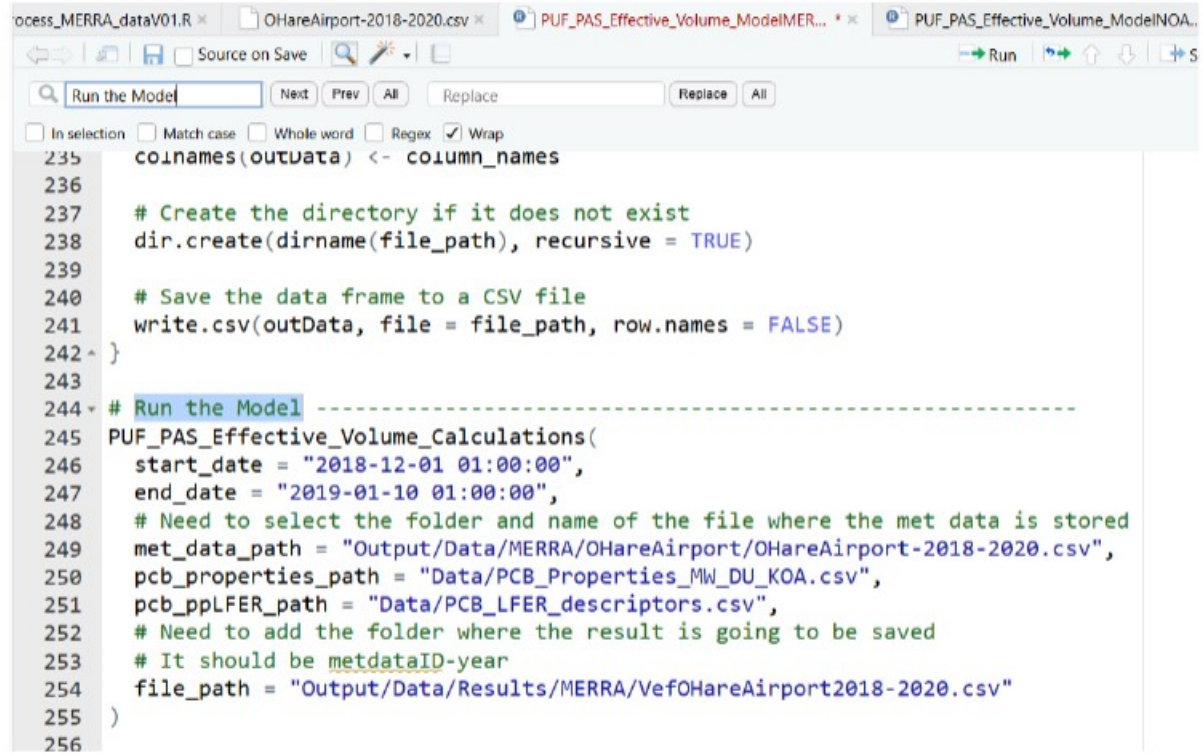
Set file_path to Output/Data/Results/GHCNh/VefChicagoOHareAirport2025.csv

The model checks exact UTC timestamps and required meteorological inputs.

MERRA Effective Volume Model Script (PUF_PAS_Effective_Volume_ModelMERRA.R)

To run the code

- No additional R packages or libraries are required
- Run the function definitions (approximately lines 6–242)
  - These functions contain the calculations needed for the effective-volume model
  - You do not need to run each individual function separately; you can select and run the entire section at once
- Go to line 244, **"Run the Model."** In this section:
  1. Enter the start and end dates of your deployment using the format **yyyy-mm-dd hh:mm:ss**. Enter 00 for minutes and seconds when they are not known or not needed. For example: 2018-12-01 01:00:00
  2. Check that "**met_data_path"** points to the correct location of your processed meteorological data (line 249)
  3. Check that **pcb_properties_path** points to the correct PCB properties file. We recommend using **"Data/PCB_Properties_MW_DU_KOA.csv"**
  4. Check that **file_path** specifies the correct location and filename for the model output, under **"Output/Data/Results/MERRA/…."**



Final model output

The results CSV contains one Veff row and one SR row for each deployment.

Deployment / Collection	UTC date-time (readable in Excel)
Length	Hourly records, including both endpoints
WS_Exceedence	Percentage of hourly wind speeds > 5 m/s
Veff	Effective sampling volume (m^3)
SR	Average sampling rate (m^3/day)
PCB columns 1-209	Congener-specific calculated values

Example: Output/Data/Results/GHCNh/VefChicagoOHareAirport2025.csv

Rerun the model for another location or deployment

- Process meteorological data with the appropriate GHCNh V02 or MERRA script.
- Change start_date and end_date to the deployment and collection times (UTC).
- Update met_data_path to the processed meteorological CSV.
- Choose a unique file_path to avoid overwriting previous results.
- Keep the PCB property paths unchanged unless using different property files.
- Run the function definitions after restarting RStudio, then run the model.

GHCNh V02: specify station ID and years; incomplete years stop at the last available observation.